

apes. And the ultimate goal of natural selection is to propagate genes, but that does not mean that the ultimate goal of people is to propagate genes. Let me show you why not.



This book is about the brain, but I will not say much about neurons, hormones, and neurotransmitters. That is because the mind is not the brain but what the brain does, and not even everything it does, such as metabolizing fat and giving off heat. The 1990s have been named the Decade of the Brain, but there will never be a Decade of the Pancreas. The brain's special status comes from a special thing the brain does, which makes us see, think, feel, choose, and act. That special thing is information processing, or computation.

Information and computation reside in patterns of data and in relations of logic that are independent of the physical medium that carries them. When you telephone your mother in another city, the message stays the same as it goes from your lips to her ears even as it physically changes its form, from vibrating air, to electricity in a wire, to charges in silicon, to flickering light in a fiber optic cable, to electromagnetic waves, and then back again in reverse order. In a similar sense, the message stays the same when she repeats it to your father at the other end of the couch after it has changed its form inside her head into a cascade of neurons firing and chemicals diffusing across synapses. Likewise, a given program can run on computers made of vacuum tubes, electromagnetic switches, transistors, integrated circuits, or well-trained pigeons, and it accomplishes the same things for the same reasons.

This insight, first expressed by the mathematician Alan Turing, the computer scientists Alan Newell, Herbert Simon, and Marvin Minsky, and the philosophers Hilary Putnam and Jerry Fodor, is now called the computational theory of mind. It is one of the great ideas in intellectual history, for it solves one of the puzzles that make up the "mind-body problem": how to connect the ethereal world of meaning and intention, the stuff of our mental lives, with a physical hunk of matter like the brain. Why did Bill get on the bus? Because he wanted to visit his grandmother and knew the bus would take him there. No other answer will do. If he hated the sight of his grandmother, or if he knew the route had changed, his body would not be on that bus. For millennia this has been

a paradox. Entities like “wanting to visit one’s grandmother” and “knowing the bus goes to Grandma’s house” are colorless, odorless, and tasteless. But at the same time they are *causes* of physical events, as potent as any billiard ball clacking into another.

The computational theory of mind resolves the paradox. It says that beliefs and desires are *information*, incarnated as configurations of symbols. The symbols are the physical states of bits of matter, like chips in a computer or neurons in the brain. They symbolize things in the world because they are triggered by those things via our sense organs, and because of what they do once they are triggered. If the bits of matter that constitute a symbol are arranged to bump into the bits of matter constituting another symbol in just the right way, the symbols corresponding to one belief can give rise to new symbols corresponding to another belief logically related to it, which can give rise to symbols corresponding to other beliefs, and so on. Eventually the bits of matter constituting a symbol bump into bits of matter connected to the muscles, and behavior happens. The computational theory of mind thus allows us to keep beliefs and desires in our explanations of behavior while planting them squarely in the physical universe. It allows meaning to cause and be caused.

The computational theory of mind is indispensable in addressing the questions we long to answer. Neuroscientists like to point out that all parts of the cerebral cortex look pretty much alike—not only the different parts of the human brain, but the brains of different animals. One could draw the conclusion that all mental activity in all animals is the same. But a better conclusion is that we cannot simply look at a patch of brain and read out the logic in the intricate pattern of connectivity that makes each part do its separate thing. In the same way that all books are physically just different combinations of the same seventy-five or so characters, and all movies are physically just different patterns of charges along the tracks of a videotape, the mammoth tangle of spaghetti of the brain may all look alike when examined strand by strand. The content of a book or a movie lies in the *pattern* of ink marks or magnetic charges, and is apparent only when the piece is read or seen. Similarly, the content of brain activity lies in the patterns of connections and patterns of activity among the neurons. Minute differences in the details of the connections may cause similar-looking brain patches to implement very different programs. Only when the program is run does the coherence become evident. As Tooby and Cosmides have written,

There are birds that migrate by the stars, bats that echolocate, bees that compute the variance of flower patches, spiders that spin webs, humans that speak, ants that farm, lions that hunt in teams, cheetahs that hunt alone, monogamous gibbons, polyandrous seahorses, polygynous gorillas. . . . There are millions of animal species on earth, each with a different set of cognitive programs. *The same basic neural tissue embodies all of these programs*, and it could support many others as well. Facts about the properties of neurons, neurotransmitters, and cellular development cannot tell you which of these millions of programs the human mind contains. Even if all neural activity is the expression of a uniform process at the cellular level, it is the arrangement of neurons—into bird song templates or web-spinning programs—that matters.

That does not imply, of course, that the brain is irrelevant to understanding the mind! Programs are assemblies of simple information-processing units—tiny circuits that can add, match a pattern, turn on some other circuit, or do other elementary logical and mathematical operations. What those microcircuits can do depends only on what they are made of. Circuits made from neurons cannot do exactly the same things as circuits made from silicon, and vice versa. For example, a silicon circuit is faster than a neural circuit, but a neural circuit can match a larger pattern than a silicon one. These differences ripple up through the programs built from the circuits and affect how quickly and easily the programs do various things, even if they do not determine exactly which things they do. My point is not that prodding brain tissue is irrelevant to understanding the mind, only that it is not enough. Psychology, the analysis of mental software, will have to burrow a considerable way into the mountain before meeting the neurobiologists tunneling through from the other side.

The computational theory of mind is not the same thing as the despised “computer metaphor.” As many critics have pointed out, computers are serial, doing one thing at a time; brains are parallel, doing millions of things at once. Computers are fast; brains are slow. Computer parts are reliable; brain parts are noisy. Computers have a limited number of connections; brains have trillions. Computers are assembled according to a blueprint; brains must assemble themselves. Yes, and computers come in putty-colored boxes and have AUTOEXEC.BAT files and run screen-savers with flying toasters, and brains do not. The claim is not that the brain is like commercially available computers. Rather, the claim is that brains and computers embody intelligence for some of the same

reasons. To explain how birds fly, we invoke principles of lift and drag and fluid mechanics that also explain how airplanes fly. That does not commit us to an Airplane Metaphor for birds, complete with jet engines and complimentary beverage service.

Without the computational theory, it is impossible to make sense of the evolution of the mind. Most intellectuals think that the human mind must somehow have escaped the evolutionary process. Evolution, they think, can fabricate only stupid instincts and fixed action patterns: a sex drive, an aggression urge, a territorial imperative, hens sitting on eggs and ducklings following hulks. Human behavior is too subtle and flexible to be a product of evolution, they think; it must come from somewhere else—from, say, “culture.” But if evolution equipped us not with irresistible urges and rigid reflexes but with a neural computer, everything changes. A program is an intricate recipe of logical and statistical operations directed by comparisons, tests, branches, loops, and subroutines embedded in subroutines. Artificial computer programs, from the Macintosh user interface to simulations of the weather to programs that recognize speech and answer questions in English, give us a hint of the finesse and power of which computation is capable. Human thought and behavior, no matter how subtle and flexible, could be the product of a very complicated program, and that program may have been our endowment from natural selection. The typical imperative from biology is not “Thou shalt . . . ,” but “If . . . then . . . else.”



The mind, I claim, is not a single organ but a system of organs, which we can think of as psychological faculties or mental modules. The entities now commonly evoked to explain the mind—such as general intelligence, a capacity to form culture, and multipurpose learning strategies—will surely go the way of protoplasm in biology and of earth, air, fire, and water in physics. These entities are so formless, compared to the exacting phenomena they are meant to explain, that they must be granted near-magical powers. When the phenomena are put under the microscope, we discover that the complex texture of the everyday world is supported not by a single substance but by many layers of elaborate machinery. Biologists long ago replaced the concept of an all-powerful protoplasm with the concept of functionally specialized

or scientists, for that matter—predict, months in advance, the trajectories of two objects thousands of miles apart to an accuracy of inches and minutes? And do it from information that can be conveyed in a few seconds of conversation? The calculus behind this forecasting is intuitive psychology: the knowledge that I *want* to meet my friend and vice versa, and that each of us *believes* the other will be at a certain place at a certain time and *knows* a sequence of rides, hikes, and flights that will take us there. No science of mind or brain is ever likely to do better. That does not mean that the intuitive psychology of beliefs and desires is itself a science, but it suggests that scientific psychology will have to explain how a hunk of matter, such as a human being, can have beliefs and desires and how the beliefs and desires work so well.



The traditional explanation of intelligence is that human flesh is suffused with a non-material entity, the soul, usually envisioned as some kind of ghost or spirit. But the theory faces an insurmountable problem: How does the spook interact with solid matter? How does an ethereal nothing respond to flashes, pokes, and beeps and get arms and legs to move? Another problem is the overwhelming evidence that the mind is the activity of the brain. The supposedly immaterial soul, we now know, can be bisected with a knife, altered by chemicals, started or stopped by electricity, and extinguished by a sharp blow or by insufficient oxygen. Under a microscope, the brain has a breathtaking complexity of physical structure fully commensurate with the richness of the mind.

Another explanation is that mind comes from some extraordinary form of matter. Pinocchio was animated by a magical kind of wood found by Geppetto that talked, laughed, and moved on its own. Alas, no one has ever discovered such a wonder substance. At first one might think that the wonder substance is brain tissue. Darwin wrote that the brain “secretes” the mind, and recently the philosopher John Searle has argued that the physico-chemical properties of brain tissue somehow produce the mind just as breast tissue produces milk and plant tissue produces sugar. But recall that the same kinds of membranes, pores, and chemicals are found in brain tissue throughout the animal kingdom, not to mention in brain tumors and cultures in dishes. All of these globs of neural tissue have the same physico-chemical properties, but not all of

them accomplish humanlike intelligence. Of course, *something* about the tissue in the human brain is necessary for our intelligence, but the physical properties are not sufficient, just as the physical properties of bricks are not sufficient to explain architecture and the physical properties of oxide particles are not sufficient to explain music. Something in the *patterning* of neural tissue is crucial.

Intelligence has often been attributed to some kind of energy flow or force field. Orbs, luminous vapors, auras, vibrations, magnetic fields, and lines of force figure prominently in spiritualism, pseudoscience, and science-fiction kitsch. The school of Gestalt psychology tried to explain visual illusions in terms of electromagnetic force fields on the surface of the brain, but the fields were never found. Occasionally the brain surface has been described as a continuous vibrating medium that supports holograms or other wave interference patterns, but that idea, too, has not panned out. The hydraulic model, with its psychic pressure building up, bursting out, or being diverted through alternative channels, lay at the center of Freud's theory and can be found in dozens of everyday metaphors: anger welling up, letting off steam, exploding under the pressure, blowing one's stack, venting one's feelings, bottling up rage. But even the hottest emotions do not literally correspond to a buildup and discharge of energy (in the physicist's sense) somewhere in the brain. In Chapter 6 I will try to persuade you that the brain does not actually *operate* by internal pressures but *contrives them* as a negotiating tactic, like a terrorist with explosives strapped to his body.

A problem with all these ideas is that even if we *did* discover some gel or vortex or vibration or orb that spoke and plotted mischief like Gepetto's log, or that, more generally, made decisions based on rational rules and pursued a goal in the face of obstacles, we would still be faced with the mystery of *how* it accomplished those feats.

No, intelligence does not come from a special kind of spirit or matter or energy but from a different commodity, *information*. Information is a correlation between two things that is produced by a lawful process (as opposed to coming about by sheer chance). We say that the rings in a stump carry information about the age of the tree because their number correlates with the tree's age (the older the tree, the more rings it has), and the correlation is not a coincidence but is caused by the way trees grow. Correlation is a mathematical and logical concept; it is not defined in terms of the stuff that the correlated entities are made of.

Information itself is nothing special; it is found wherever causes leave

effects. What is special is information *processing*. We can regard a piece of matter that carries information about some state of affairs as a symbol; it can “stand for” that state of affairs. But as a piece of matter, it can do other things as well—physical things, whatever that kind of matter in that kind of state can do according to the laws of physics and chemistry. Tree rings carry information about age, but they also reflect light and absorb staining material. Footprints carry information about animal motions, but they also trap water and cause eddies in the wind.

Now here is an idea. Suppose one were to build a machine with parts that are affected by the physical properties of some symbol. Some lever or electric eye or tripwire or magnet is set in motion by the pigment absorbed by a tree ring, or the water trapped by a footprint, or the light reflected by a chalk mark, or the magnetic charge in a bit of oxide. And suppose that the machine then causes something to happen in some other pile of matter. It burns new marks onto a piece of wood, or stamps impressions into nearby dirt, or charges some other bit of oxide. Nothing special has happened so far; all I have described is a chain of physical events accomplished by a pointless contraption.

Here is the special step. Imagine that we now try to interpret the newly arranged piece of matter using the scheme according to which the original piece carried information. Say we *count* the newly burned wood rings and interpret them as the age of some tree at some time, even though they were not caused by the growth of any tree. And let’s say that the machine was carefully designed so that the interpretation of its new markings made sense—that is, so that they carried information about something in the world. For example, imagine a machine that scans the rings in a stump, burns one mark on a nearby plank for each ring, moves over to a smaller stump from a tree that was cut down at the same time, scans its rings, and sands off one mark in the plank for each ring. When we count the marks on the plank, we have the age of the first tree at the time that the second one was planted. We would have a kind of *rational* machine, a machine that produces true conclusions from true premises—not because of any special kind of matter or energy, or because of any part that was itself intelligent or rational. All we have is a carefully contrived chain of ordinary physical events, whose first link was a configuration of matter that carries information. Our rational machine owes its rationality to two properties glued together in the entity we call a symbol: a symbol carries information, and it causes things to happen. (Tree rings correlate with the age of the tree, and they can absorb the light beam of a scanner.)

When the caused things themselves carry information, we call the whole system an information processor, or a computer.

Now, this whole scheme might seem like an unrealizable hope. What guarantee is there that *any* collection of thingamabobs can be arranged to fall or swing or shine in just the right pattern so that when their effects are interpreted, the interpretation will make sense? (More precisely, so that it will make sense according to some prior law or relationship we find interesting; any heap of stuff can be given a contrived interpretation after the fact.) How confident can we be that some machine will make marks that actually correspond to some meaningful state of the world, like the age of a tree when another tree was planted, or the average age of the tree's offspring, or anything else, as opposed to being a meaningless pattern corresponding to nothing at all?

The guarantee comes from the work of the mathematician Alan Turing. He designed a hypothetical machine whose input symbols and output symbols could correspond, depending on the details of the machine, to any one of a vast number of sensible interpretations. The machine consists of a tape divided into squares, a read-write head that can print or read a symbol on a square and move the tape in either direction, a pointer that can point to a fixed number of tickmarks on the machine, and a set of mechanical reflexes. Each reflex is triggered by the symbol being read and the current position of the pointer, and it prints a symbol on the tape, moves the tape, and/or shifts the pointer. The machine is allowed as much tape as it needs. This design is called a Turing machine.

What can this simple machine do? It can take in symbols standing for a number or a set of numbers, and print out symbols standing for new numbers that are the corresponding value for any mathematical function that can be solved by a step-by-step sequence of operations (addition, multiplication, exponentiation, factoring, and so on—I am being imprecise to convey the importance of Turing's discovery without the technicalities). It can apply the rules of any useful logical system to derive true statements from other true statements. It can apply the rules of any grammar to derive well-formed sentences. The equivalence among Turing machines, calculable mathematical functions, logics, and grammars, led the logician Alonzo Church to conjecture that *any* well-defined recipe or set of steps that is guaranteed to produce the solution to some problem in a finite amount of time (that is, any algorithm) can be implemented on a Turing machine.

What does this mean? It means that to the extent that the world

obeys mathematical equations that can be solved step by step, a machine can be built that simulates the world and makes predictions about it. To the extent that rational thought corresponds to the rules of logic, a machine can be built that carries out rational thought. To the extent that a language can be captured by a set of grammatical rules, a machine can be built that produces grammatical sentences. To the extent that thought consists of applying *any* set of well-specified rules, a machine can be built that, in some sense, thinks.

Turing showed that rational machines—machines that use the physical properties of symbols to crank out new symbols that make some kind of sense—are buildable, indeed, easily buildable. The computer scientist Joseph Weizenbaum once showed how to build one out of a die, some rocks, and a roll of toilet paper. In fact, one doesn't even need a huge warehouse of these machines, one to do sums, another to do square roots, a third to print English sentences, and so on. One kind of Turing machine is called a universal Turing machine. It can take in a *description* of any other Turing machine printed on its tape and thereafter mimic that machine exactly. A single machine can be programmed to do anything that any set of rules can do.

Does this mean that the human brain is a Turing machine? Certainly not. There are no Turing machines in use anywhere, let alone in our heads. They are useless in practice: too clumsy, too hard to program, too big, and too slow. But it does not matter. Turing merely wanted to prove that *some* arrangement of gadgets could function as an intelligent symbol-processor. Not long after his discovery, more practical symbol-processors were designed, some of which became IBM and Univac mainframes and, later, Macintoshes and PCs. But all of them were equivalent to Turing's universal machine. If we ignore size and speed, and give them as much memory storage as they need, we can program them to produce the same outputs in response to the same inputs.

Still other kinds of symbol-processors have been proposed as models of the human mind. These models are often simulated on commercial computers, but that is just a convenience. The commercial computer is first programmed to emulate the hypothetical mental computer (creating what computer scientists call a virtual machine), in much the same way that a Macintosh can be programmed to emulate a PC. Only the virtual mental computer is taken seriously, not the silicon chips that emulate it. Then a program that is meant to model some sort of thinking (solving a problem, understanding a sentence) is run on the virtual men-

tal computer. A new way of understanding human intelligence has been born.



Let me show you how one of these models works. In an age when real computers are so sophisticated that they are almost as incomprehensible to laypeople as minds are, it is enlightening to see an example of computation in slow motion. Only then can one appreciate how simple devices can be wired together to make a symbol-processor that shows real intelligence. A lurching Turing machine is a poor advertisement for the theory that the mind is a computer, so I will use a model with at least a vague claim to resembling our mental computer. I'll show you how it solves a problem from everyday life—kinship relations—that is complex enough that we can be impressed when a machine solves it.

The model we'll use is called a production system. It eliminates the feature of commercial computers that is most starkly unbiological: the ordered list of programming steps that the computer follows single-mindedly, one after another. A production system contains a memory and a set of reflexes, sometimes called "demons" because they are simple, self-contained entities that sit around waiting to spring into action. The memory is like a bulletin board on which notices are posted. Each demon is a knee-jerk reflex that waits for a particular notice on the board and responds by posting a notice of its own. The demons collectively constitute a program. As they are triggered by notices on the memory board and post notices of their own, in turn triggering other demons, and so on, the information in memory changes and eventually contains the correct output for a given input. Some demons are connected to sense organs and are triggered by information in the world rather than information in memory. Others are connected to appendages and respond by moving the appendages rather than by posting more messages in memory.

Suppose your long-term memory contains knowledge of the immediate families of you and everyone around you. The content of that knowledge is a set of propositions like "Alex is the father of Andrew." According to the computational theory of mind, that information is embodied in symbols: a collection of physical marks that correlate with the state of the world as it is captured in the propositions.

These symbols cannot be English words and sentences, notwith-